Project 1: Khai thác mẫu có trọng số tích hợp ngưỡng động

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 2: Khai thác Top-k mẫu có trọng số

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 3: Khai thác mẫu có mục tiêu

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 4: Khai thác mẫu có mục tiêu không gian - thời gian

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 5: Áp dụng vector embedding trong phân cụm dữ liệu chuỗi có trọng số

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 6: Sequence clustering algorithm based on weighted vector identification

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 7: Temporal Data Clustering via Weighted Clustering Ensemble with Different Representations

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 8: Áp dụng TDA và CNN cho bài toán phân lớp dữ liệu ảnh/video

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 9: Áp dụng TDA và transformer cho bài toán phân loại tin tức giả mạo trên mạng xã hội

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 10: FakeShield: Explainable Image Forgery Detection and Localization via Multi-modal Large Language Models

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 11: Personalized Hotel Recommender System

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 12: Restaurant Recommendation System using Machine Learning

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 13: Fake review detection system for online E-commerce platforms

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 14: Phishing Website Detection Using Deep Learning Models

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 15: Automatic phishing website detection and prevention model using transformer deep belief network

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 16: Deep learning-based anomaly detection in real-time video

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 17: Prediction of electricity consumption using an innovative deep energy predictor model for enhanced accuracy and efficiency

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 18: Anomaly detection using transfer learning for electricity consumption

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 19: Pneumonia Detection Based on Chest X-Ray Images with MobileNetV2 Architecture

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 20: A web-based pneumonia detection system using deep learning on chest X-ray image

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 21: Segment anything in medical images

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 22: DNA Sequence Classification Using RRCNN and LSTM

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 23: Pattern Discovery in Genomic Sequences Using Data Mining Algorithms

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 24: Explainable AI for Parkinson's disease prediction: A machine learning approach with interpretable models

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 25: Time Series Prediction Based on LSTM-Attention-LSTM Model

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 26: SegmentAnyBone: A universal model that segments any bone at any location on MRI

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 27: Software bug prediction using graph neural networks and graph-based text representations

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 28: NerdBug: automated bug detection in neural networks

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 29: Hierarchical convolutional neural networks for fashion image classification

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 30: Algorithm Analysis of Clothing Classification Based on Neural Network

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 31: Image-Based Virtual Try-On

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 32: Personalised Fashion Recommendation using Deep Learning

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 33: Analyzing customer support calls

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 34: Certs authentication system using blockchain

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 35: Customer Churn Analysis Prediction

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 36: Credit Card Fraud Detection using Machine Learning Algorithms

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 37: Transaction Fraud Detection via an Adaptive Graph Neural Network

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 38: Detecting Fraud Apps using Sentiment Analysis

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 39: Online Hospital Appointment Booking

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 40: Design and Implementation of an Online Blood Donation System

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 41: Medical Report Management & Distribution System on Blockchain

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 42: Predicting stroke risk: An effective stroke prediction model based on neural networks

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 43: Toxic Comment Classification System Using Deep Learning

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 44: Pharma Supply Chain System using Smart Contracts

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 45: Secure pharmaceutical supply chain using blockchain in IoT cloud systems

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 46: Android-Based Image Steganography System

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 47: Online electronic Voting Systems

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 48: Blockchain Shipment Management Tracking System

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 49: A blockchain-based approach to smart cargo transportation using UHF RFID

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 50: Secured Tracking and Tracing System Based on Blockchain Technology

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 51: Vaccination System using Blockchain Technology

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 52: Real Estate Management System using Blockchain

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 53: Trusted Crowdfunding Platform Using a Smart Contract

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 54: Time series anomaly detection via temporal relationship graphs and adaptive smoothing

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 55: Leaf disease detection using machine learning

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 56: LoC — A new financial loan management system based on smart contracts

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 57: A Blockchain-Based Mobile Payment System for Secure Transaction

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 58: Smart blockchain networks: Revolutionizing donation tracking in the Web 3.0

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 59: Smart EV Charger Locator and Slot Booking Manager App

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 60: Image Based Password Authentication System

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 61: Online Fashion Stylist System

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 62: Detecting E Banking Phishing Websites Using Associative Classification

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 63: Real-time Vehicle Tracking Using Driver Mobile Gps Tracking

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 64: An image encryption scheme based on a modified AES algorithm by using a variable S-box

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 65: Image Encryption using Reverse Data Hiding Algorithm with Triple DES

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 66: Image Steganography using 3 ways of Encryption for Highly Secured Data Transmission

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 67: Secure Data Transfer Over Internet Using Image Steganography

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 68: Customer Behaviour Prediction for E-Commerce Sites Using Web Mining and Machine Learning

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.

Project 69: Enhanced ATM Security with OTP Based Authentication

1. Introduction

This project investigates modern techniques in artificial intelligence, data mining, or software systems to solve real-world problems. The goal is to design and implement a practical solution supported by recent research and experimental evaluation.

2. Objectives

• Identify the research problem and design a technical solution.
• Develop algorithms or intelligent models.
• Implement a working prototype system.
• Evaluate performance using datasets and metrics.

3. System Architecture

The system typically consists of modules for data collection, preprocessing, model development, evaluation, and user interface or deployment environment.

4. Methodology

1. Data collection and preprocessing
2. Feature extraction or representation learning
3. Model development using machine learning / deep learning
4. Experimental evaluation
5. Deployment or visualization

5. Input / Output

Input: Dataset relevant to the problem domain.
Output: Predictions, detected patterns, recommendations, or analytical results.

6. Tools and Technologies

Python, PyTorch/TensorFlow, Scikit-learn, databases, visualization libraries, and cloud or web frameworks depending on the project.

7. Impact and Applications

Applications may include healthcare analytics, cybersecurity, recommender systems, smart cities, financial technology, or industrial automation.

8. References (2020–2025)

Recent research papers from IEEE, ACM, Elsevier, Springer, and major AI conferences such as NeurIPS, ICML, and CVPR.